

Apply filters to SQL queries

Scenario

You are a security professional at a large organization. Part of your job is to investigate security issues to help keep the system secure. You recently discovered some potential security issues that involve login attempts and employee machines.

Your task is to examine the organization's data in their employees and log_in_attempts tables. You'll need to use SQL filters to retrieve records from different datasets and investigate the potential security issues.

Project description

As a security professional at a large organization, I am responsible for investigating potential security issues, strengthening system defenses, and ensuring employee machines remain up to date. In this project, I utilized SQL to extract meaningful data from the log_in_attempts and employees tables by applying filters to investigate suspicious login activity and identify employee devices needing updates. The queries below demonstrate how I applied SQL filters using operators such as AND, OR, and NOT, along with pattern matching using LIKE.

Retrieve after-hours failed login attempts.

I wrote a SQL query to filter all login attempts that failed and occurred after 6:00 PM. This involved selecting records from the log_in_attempts table where the login_time was greater than '18:00' and the success field was FALSE, which filters for the failed login attempts.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_time > '18:00' AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0

Retrieve login attempts on specific dates.

To find relevant login attempts, I queried the log_in_attempts table for all records where the login_date was either '2022-05-09' or '2022-05-08'. The first condition is login_date = '2022-05-09', which filters for logins on 2022-05-09. The second condition is login_date = '2022-05-08', which filters for logins on 2022-05-08.

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

Retrieve login attempts outside of Mexico.

I started by selecting all data from the `log_in_attempts` table. Then, I used a `WHERE` clause with `NOT` To filter for countries other than Mexico. I used `LIKE` with `MEX%` as the pattern to match because the dataset represents Mexico as `MEX` and `MEXICO`. The wildcard `%` was used to capture variations in how Mexico may be recorded.

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1

Retrieve employees in marketing.

I started by selecting all data from the `employees` table. Then, I used a `WHERE` clause with `AND` to filter for employees who work in the Marketing department and in the East building. I queried the `employees` table to return records for those in the Marketing department and whose office is in the East wing. I used `LIKE 'East%'` to account for various East office numbers.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

Retrieve employees in finance or sales.

I started by selecting all data from the employees table. Then, I used a WHERE clause with OR to filter for employees who are in the Finance and Sales departments. I used the OR operator instead of AND because I want all employees who are in either department. The first condition is department = 'Finance', which filters for employees from the Finance department. The second condition is department = 'Sales', which filters for employees from the Sales department.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170

Retrieve all employees not in IT.

First, I started by selecting all data from the employees table. Then, I used a WHERE clause with NOT to filter for employees not in this department. I filtered the employees table using the NOT operator to exclude records where the department was 'Information Technology'.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434

Summary

In this project, I applied SQL filters to investigate system security issues and identify target employee groups for updates. By querying the log_in_attempts and employees tables, I retrieved specific datasets using logical operators (AND, OR, NOT) and pattern matching (LIKE, %). These queries supported our security team in identifying abnormal activity and ensuring timely system maintenance for employees.